



MUFFAKHAM JAH COLLEGE OF ENGINEERING & TECHNOLOGY

(SULTAN-UL-ULOOM EDUCATION SOCIETY)

COMPUTER SCIENCE AND ENGINEERING DEPARTMENT

**LABORATORY EXERCISE PLAN FOR SCIENTIFIC PROGRAMMING LAB**

**BE II SEM, ACADEMIC YEAR 2024 – 25**

**Name of Faculty: Ms. Masrath Parveen**

**Branch/DIV :CSE 3 & 6**

| Unit No  | Program                                                                  | No of Hours       |
|----------|--------------------------------------------------------------------------|-------------------|
| Unit - 1 | Python and MATLAB Installation and IDE setup.                            | 2 weeks - 4 hours |
|          | Program to convert temperatures to and from Celsius, Fahrenheit.         |                   |
|          | Program to perform different Arithmetic Operations on numbers in Python. |                   |
|          | Write a python program to check if a given number is even or odd.        |                   |
|          | Write a python program to find largest of three numbers.                 |                   |
|          | Program to print overall grade of a student in python.                   |                   |
|          | Program to print 3 numbers in ascending order.                           |                   |
|          | Program to find factorial of a given number.                             |                   |
|          | Program to print natural numbers and find the sum up to given limit.     |                   |
|          | Program to print Fibonacci series up to a given limit.                   |                   |
|          | Print multiplication table of a given number.                            |                   |
|          | Print multiplication tables up to given limit.                           |                   |
|          | Program to check if a given number is prime or not.                      |                   |
| unit - 2 | Write a program to demonstrate working with strings in python.           | 3 weeks - 6 hours |
|          | Write a program to create, append, and remove lists in python.           |                   |
|          | Write a program to demonstrate working with tuples in python.            |                   |
|          | Write a program to demonstrate working with dictionaries in python.      |                   |
|          | Stack implementation in python.                                          |                   |
|          | Queue implementation in python.                                          |                   |
|          | Linear and Binary search programs in python.                             |                   |
| Unit - 3 | Selection and Bubble sort programs in python.                            | 2 weeks - 4 hours |
|          | Finding roots using Newton- Raphson and Bisection methods in python.     |                   |
|          | Trapezoidal Rule (Numerical Integration) program in python.              |                   |
|          | Performing Matrix Operations and Solving Linear equations                |                   |
| Unit - 4 | Finding the Determinant and Rank of a Matrix.                            | 2 weeks - 4 hours |
|          | Data Manipulation and Analysis using Pandas.                             |                   |
|          | Generating Various Plots(Line, Scatter, Bar, Histogram)                  |                   |
|          | Customizing and Interpreting Plots for Engineering Data.                 |                   |
|          | Linear Regression Example (Supervised Learning)                          |                   |

*Masrath Parveen*

IN-CHARGE

*Shabbir*

HEAD CSED